Autonomous Deep-Sea Forensic Data Rescue System Using Embedded Systems and Acoustic Telemetry

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Primary Discipline: Digital Forensics, Embedded Systems, Cyber-Physical Security

Abstract Digital evidence and forensic storage media are highly susceptible to total destruction in abyssal ocean environments due to extreme hydrostatic pressure, saltwater corrosion, and physical impacts. Traditional protective housings often rely on passive mechanical seals and unmonitored recovery methods, leading to structural fatigue and irreversible loss of data integrity. This paper presents a specialized autonomous deep-sea forensic capsule designed to encapsulate micro-SSD media within a Grade-5 Titanium alloy body. The system utilizes an internal fluid-filled environment (fluorinated liquid/mineral oil) to equalize electrical pressure and ensure positive buoyancy, coupled with a nitrogen-gas hydraulic ejection piston mechanism controlled by an ESP32 microcontroller and a fail-safe solenoid latch. For precise recovery, a hybrid acoustic telemetry system transmits positional telemetry through underwater acoustic modems. This hybrid approach guarantees data integrity, mitigates mechanical fatigue, and provides an end-to-end framework for digital evidence preservation in abyssal forensic investigations.

1. Introduction & Forensic Problem Statement In maritime disasters, aviation crashes over oceanic basins, or naval forensic operations, digital evidence stored on solid-state drives (SSDs) is subject to extreme abyssal conditions. Submersion at significant depths exposes storage devices to: Mechanical Fatigue & Crushing: Extreme hydrostatic pressure deforms conventional protective enclosures. Corrosive Degradation: Saltwater ingress corrodes flash memory controllers and NAND arrays. Loss of Forensic Integrity: Physical damage alters the hash values (MD5/SHA-256) of critical file system artifacts, rendering evidence inadmissible. To prevent irreversible data loss, there is an urgent need for an active, self-

ejecting, pressure-resistant capsule that insulates digital media while autonomously ascending to accessible depths for forensic recovery.

2. Mechanical Design & System Architecture

2.1 Housing & Structural Insulation Chamber Material: Grade-5 Titanium Alloy (Ti-6Al-4V), selected for its high strength-to-weight ratio and exceptional resistance to marine corrosion. Internal Fluid Dielectric: The inner cavity is filled with non-conductive dielectric fluid (mineral oil / fluorinated liquid) to absorb shock, insulate electronics, and eliminate internal air gaps that cause structural collapse under pressure. Marine Shock Absorption: Sealed using high-grade marine epoxy and specialized elastomer seals instead of brittle ceramic fusion to maintain structural ductility during high-impact events.

2.2 Ejection Mechanism (Nitrogen Hydraulic Pistons) Unlike traditional mechanical springs—which suffer from severe metal fatigue and degradation under abyssal pressure—this system employs pressurized Nitrogen-Gas Hydraulic Pistons. Trigger Mechanism: An ESP32 microcontroller interfaces with an array of environmental sensors (pressure, liquid ingress, timer thresholds) and controls a Fail-Safe Solenoid Latch. Deployment: Upon triggering or power interruption, the solenoid releases the latch, allowing nitrogen expansion to drive the hydraulic piston, mechanically decoupling the inner core and propelling the capsule toward the surface using positive liquid buoyancy. 3. Forensic Integrity & Data Recovery Protocol

3.1 Hardware-Level Data Preservation By preventing physical contact between seawater and the storage substrate (micro-SSD), the system preserves: Uncorrupted partition tables (GPT/MBR) and raw file systems (NTFS, ext4, APFS). Volatile memory/flash artifacts essential for timeline reconstruction.

3.2 Hash Matching & Chain of Custody The design ensures that upon retrieval, forensic examiners can perform standard bit-stream disk imaging (e.g., using Autopsy, FTK Imager, or DD utilities) without requiring destructive physical restoration. The preserved hash value matches the pre-impact cryptographic signature, adhering strictly to ISO/IEC 27037 standards for digital evidence handling.

4. Acoustic Telemetry & Recovery Mechanics Underwater Acoustic Modem: Because high-frequency radio waves (Wi-Fi, GPS, Cellular) attenuate rapidly in seawater, the capsule utilizes low-frequency Acoustic Telemetry (Ultrasonic/Acoustic Modems) for underwater tracking. Surface Beacon: Once the capsule ejects and breaches the sea surface, a dual-mode system activates: an acoustic pinger for short-range submerged recovery and a satellite positioning beacon for surface geolocation. 5. Conclusion The proposed system addresses a critical gap in maritime digital forensics. By unifying advanced mechanical

resilience (Grade-5 Titanium & Nitrogen Hydraulic Pistons) with embedded micro-controllers and acoustic location tracking, the architecture guarantees the physical and cryptographic integrity of digital evidence extracted from extreme marine depths.

Keywords Digital Forensics, Abyssal Data Recovery, Grade-5 Titanium Capsule, Nitrogen Hydraulic Ejection, Acoustic Telemetry, Embedded Systems, Evidence Integrity.

